Algebraic Limits: Factoring and Rationalizing

Answer Key

AP Calculus AB/BC

Quick Answers

1. 6
2. -1
3. 12
4. $\frac{1}{4}$
5. $\frac{1}{6}$
6. $-\frac{1}{4}$
7. 9
8. —

Curriculum

Level: AP Calculus AB/BC
LIM-1 – *problems 1, 2, 3, 4, 5, 6*
LIM-2 – *problems 7, 8*
Difficulty 1–5 of 5 across 8 tagged checks.

1. Direct substitution gives $\frac{0}{0}$. Factor the numerator as a difference of squares: $\dfrac{x^2-9}{x-3} = \dfrac{(x-3)(x+3)}{x-3} = x+3$ for $x \neq 3$. Substituting $x=3$ gives $\boxed{6}$, which matches the table's approach from both sides.

2. Factor: $x^2-5x+6=(x-2)(x-3)$, so the quotient is $x-3$ for $x \neq 2$. Substituting $x=2$ gives $\boxed{-1}$.

3. Difference of cubes: $x^3-8=(x-2)(x^2+2x+4)$, so the quotient is x^2+2x+4. At $x=2$ this is $4+4+4=\boxed{12}$.

4. Multiply by the conjugate $\dfrac{\sqrt{x+4}+2}{\sqrt{x+4}+2}$. The numerator becomes $(x+4)-4=x$, giving $\dfrac{x}{x(\sqrt{x+4}+2)} = \dfrac{1}{\sqrt{x+4}+2}$. At $x=0$ this is $\dfrac{1}{2+2}=\boxed{1/4}$.

5. Multiply by $\dfrac{\sqrt{x}+3}{\sqrt{x}+3}$. The numerator becomes $x-9$, so the quotient is $\dfrac{1}{\sqrt{x}+3}$. At $x=9$ this is $\dfrac{1}{3+3}=\boxed{1/6}$.

6. Multiply by $\dfrac{2+\sqrt{x+3}}{2+\sqrt{x+3}}$. The numerator becomes $4-(x+3)=1-x=-(x-1)$, so the quotient is $\dfrac{-1}{2+\sqrt{x+3}}$. At $x=1$ this is $\dfrac{-1}{2+2}=\boxed{-1/4}$. The negative sign comes from the radical being subtracted, not from an algebra slip.

7. For $x \neq 7$, $\dfrac{x^2 - 5x - 14}{x - 7} = \dfrac{(x-7)(x+2)}{x-7} = x + 2$, so $\lim_{x \to 7} f(x) = 9$. Continuity at $x = 7$ requires $f(7)$ to equal that limit, so $c = \boxed{9}$.

8. **Manual review — expected reasoning.** A limit describes the behaviour of f as x approaches 3, and the definition of the limit never evaluates f at 3 itself. For every $x \neq 3$ the two expressions agree, so they have the same limit. At $x = 3$ they differ: $x + 3$ is defined and equals 6, while $\dfrac{x^2 - 9}{x - 3}$ is undefined — a removable discontinuity. Accept any answer making both points: agreement away from the point, and the limit not depending on the value at the point. Do not accept "the 3s cancel."